Test Automation Framework (TAF::Perl)

TAF Quick Start

MariaDB Foundation https://mariadb.org

Open, vendor-neutral, community-driven tooling for database testing and benchmarking.

TAF-Perl is maintained and published by the MariaDB Foundation to support transparent, reproducible, and fair performance evaluation across the open source database ecosystem. The framework is part of the Foundation's mission to provide open tooling that benefits the entire database community.

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TAF QUICK START GUIDE
Test Automation Framework (TAF) for Database Software Installs and Workloads
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This Quick Start is for users who want to run TAF immediately and read the details later. The goal is simple: install a database, build the required client (sysbench-lua), run tests, and collect results.

Note: There are other guides and overview pdfs in: taf-perl/help/

TAF supports two primary install paths:

1. Action-based installs (fully automated)
2. Direct command-line installs (manual install, then run tests)

This guide provides four examples:
- MariaDB: full action install
- MariaDB: command-line install
- MySQL: full action install
- MySQL: command-line install

Only one client currently requires building: sysbench-lua.

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REQUIREMENTS
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* Linux-x86
* A database software package (tarball or RPM bundle) accessible on local host.
* Perl 5

Note: Examples are from RedHat/OracleLinux 8
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KEY OPTIONS
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--db-software-install
  Perform a direct database software install without requiring a properties file or TAF action.

--db-software-install-packages=/path/pkg1.tar.gz[,pkg2.rpm,...]
  Packages to install. Required for both action-based and direct installs.

--action=<action-name>
  Execute a predefined TAF action. Examples include:
    install-init-db-exit
    install-init-start-db-exit
    install-init-start-db-run-tests
    install-init-start-db-build-client-run-tests

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==
QUICK START EXAMPLE 1 - MariaDB Full Action Install
Install -> Init -> Start -> Build sysbench-lua -> Run Tests
==

This example uses the action:

  --action=install-init-start-db-build-client-run-tests

TAF will automatically:
  1. Unpack and install MariaDB
  2. Initialize the database
  3. Start the runtime server
  4. Build sysbench-lua against the installed MariaDB
  5. Run the sysbench tests defined in the properties file
  6. Archive results and logs
  7. Shut down cleanly

COMMAND
----

perl ./taf.pl \
  --prop=./properties/**mariadb**/beta/sysbench_lua.properties \
  --action=install-init-start-db-build-client-run-tests \
  --db-software-install-packages=/opt/packages/**mariadb**-10.11.16-rhel8.tar \
  --verbose

EXPECTED OUTPUT (CURATED AND MINIMAL)

Framework   : taf-perl
Framework Version   : 2.0
Run Log initialized at : /opt/taf/logs/run.log

TAF ACTION: install-init-start-db-build-client-run-tests
Base package detected: /opt/packages/mariadb-10.11.16-rhel8.tar

Install directory   : /opt/taf/database_software_installs/mariadb-10.11.16
Active install marker : /opt/taf/database_software_installs/mariadb-10.11.16
Install and Active Marker match; treating as implicit

Selected library path: /opt/taf/database_software_installs/mariadb-10.11.16/lib64
LD_LIBRARY_PATH set to /opt/taf/database_software_installs/mariadb-10.11.16/lib64

Loading database plugin: mariadb
Database plugin loaded and ready

MariaDB -> Init Database -> Complete
MariaDB -> Database Start -> Server is ready

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SYSBENCH-LUA CLIENT BUILD

Detected maker from install_dir -> mariadb
Using config tool paths:
  INC = /opt/taf/database_software_installs/mariadb-10.11.16/include/mysql
  LIB = /opt/taf/database_software_installs/mariadb-10.11.16/lib64
Resolved libmysqlclient = /opt/taf/database_software_installs/mariadb-10.11.16/lib64/libmysqlclient.so
Linux Sysbench Building!

SYSBENCH PREPARE AND RUN

Running sysbench prepare:
sysbench oltp_point_select.lua --db-driver=mysql --mysql-socket=/opt/taf/tmp/db.sock ...

POINT_SELECT -> Threads: 2 -> Iter: 1
Test execution complete

REPORTING AND ARCHIVING

Results archived to: /opt/taf/archive/HOST_POINT_SELECT_2026_02_09_113915/

CLEAN SHUTDOWN

MariaDB -> Database Stop -> Server stopped cleanly
TAF Exit Code: 0
No errors have been recorded.

ACTIVE INSTALL CHECK
----

perl taf.pl --list-database-software-installs

[ACTIVE] 1: /opt/taf/database_software_installs/mariadb-10.11.16

Once the install is active, any test suite can be run directly as long as they use the current db maker:

perl taf.pl --prop=./properties/**mariadb**/beta/sysbench_lua.properties
perl taf.pl --prop=./properties/**mariadb**/beta/hammerdb_tprocc_beta.properties
perl taf.pl --prop=./properties/**mariadb**/beta/hammerdb_tproch_beta.properties

==
END OF EXAMPLE 1
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QUICK START EXAMPLE 2 - MariaDB Command-Line Install
Install -> Mark Active -> Run Tests Against Active Install
==

This example shows how to install MariaDB using the direct command-line installer instead of a TAF action.

This method only installs the software; it does NOT init, start, or run tests automatically.

NOTE:

If you already ran Example #1, the sysbench-lua client is built and ready.

If not, you must build the client before running sysbench-lua tests.

Available client build actions:

```
"build-client"                          => Build test suite client
"build-client-run-tests"                => Build client and run tests
"init-start-db-build-client-run-tests" => Init and start DB, build client, run tests
"start-db-build-client-run-tests"       => Start DB, build client, run tests
```

COMMAND

```
perl ./taf.pl \
  --db-software-install \
  --db-software-install-packages=/opt/packages/mariadb-11.8.6-linux-systemd-x86_64.tar.gz
```

EXPECTED OUTPUT (CURATED AND MINIMAL)

```
Created staging dir: /tmp/taf_unpack_xxxxx
Unpacking base package: /opt/packages/mariadb-11.8.6-linux-systemd-x86_64.tar.gz
Unified install root = /tmp/taf_unpack_xxxxx/install_root

Moving staged install to final dir:
  /opt/taf/database_software_installs/mariadb-11.8.6-linux-systemd-x86_64

Active install set to:
  /opt/taf/database_software_installs/mariadb-11.8.6-linux-systemd-x86_64

Install has completed. Happy benchmarking!
```

VERIFY ACTIVE INSTALL

```
perl ./taf.pl --list-database-software-installs

== AVAILABLE DATABASE SOFTWARE INSTALLS ====================

  1: /opt/taf/database_software_installs/mariadb-10.11.16
[ACTIVE] 2: /opt/taf/database_software_installs/mariadb-11.8.6-linux-systemd-x86_64
```

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RUNNING TESTS AGAINST THE ACTIVE INSTALL

Once the install is active, any test suite can be run directly as long as they use the current db maker:

perl taf.pl --prop=./properties/**mariadb**/beta/sysbench_lua.properties
perl taf.pl --prop=./properties/**mariadb**/beta/hammerdb_tprocc_beta.properties
perl taf.pl --prop=./properties/**mariadb**/beta/hammerdb_tproch_beta.properties

----

END OF EXAMPLE 2

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QUICK START EXAMPLE 3 - MySQL Full Action Install
Install -> Init -> Start -> Build sysbench-lua -> Run Tests

==

This example mirrors the MariaDB full-action workflow, but uses a MySQL RPM bundle.

TAF automatically installs MySQL, initializes it, starts it, builds sysbench-lua, runs the tests, and archives results.

COMMAND

```
perl ./taf.pl \
  --prop=./properties/mysql/beta/sysbench_lua.properties \
  --action=install-init-start-db-build-client-run-tests \
  --db-software-install-packages=/opt/packages/mysql-9.6.0-el8.rpm-bundle.tar \
  --verbose
```

EXPECTED OUTPUT (CURATED AND MINIMAL)

```
Framework            : taf-perl
Framework Version    : 2.0
Run Log initialized at : /opt/taf/logs/run.log

TAF ACTION: install-init-start-db-build-client-run-tests
Base package detected: /opt/packages/mysql-9.6.0-el8.rpm-bundle.tar

Created staging dir: /tmp/taf_unpack_xxxxx
Unpacking MySQL RPM bundle...
Moving staged install to final dir:
  /opt/taf/database_software_installs/mysql-9.6.0-el8.rpm-bundle

Active install set to:
  /opt/taf/database_software_installs/mysql-9.6.0-el8.rpm-bundle
```

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MYSQL INITIALIZATION AND STARTUP

Loading database plugin: mysql
Database plugin loaded and ready

MySQL -> Running --initialize
MySQL -> Bootstrap mysqld is ready
MySQL -> Setup Users -> root password set
MySQL -> Setup Users -> tester user created
MySQL -> BootstrapStop -> Complete

MySQL -> Start -> mysqld runtime server ready
Database started successfully.

SYSBENCH-LUA CLIENT BUILD

Detected maker from install_dir -> mysql
Config tool unusable: mysql_config
Falling back to mysql_config-64

Using config tool paths:
  INC = /opt/taf/database_software_installs/mysql-9.6.0/include/mysql
  LIB = /opt/taf/database_software_installs/mysql-9.6.0/lib64/mysql

Resolved libmysqlclient = /opt/taf/database_software_installs/mysql-9.6.0/lib64/mysql/libmysqlclient.so
Linux Sysbench Building!

SYSBENCH PREPARE AND RUN

Running sysbench prepare:
sysbench oltp_point_select.lua --db-driver=mysql --mysql-socket=/opt/taf/tmp/db.sock ...

POINT_SELECT -> Threads: 2 -> Iter: 1
Test execution complete

REPORTING AND ARCHIVING

Reports archived to:
  /opt/taf/archive/HOST_POINT_SELECT_install-init-start-db-build-client-run-tests_2026_02_09_140417/

CLEAN SHUTDOWN

MySQL -> Database Stop -> Server stopped cleanly
TAF Exit Code: 0
No errors have been recorded.

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VERIFY ACTIVE INSTALL (IMPORTANT)
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perl ./taf.pl --list-database-software-installs

== AVAILABLE DATABASE SOFTWARE INSTALLS ===================

  1: /opt/taf/database_software_installs/mariadb-10.11.16
  2: /opt/taf/database_software_installs/mariadb-11.8.6
[ACTIVE] 3: /opt/taf/database_software_installs/mysql-9.6.0-el8.rpm-bundle

Once the install is active, any test suite can be run directly as long as they use the current db maker:

perl taf.pl --prop=./properties/**mysql**/beta/sysbench_lua.properties
perl taf.pl –prop=./properties/**mysql**/beta/hammerdb_tprocc_beta.properties
perl taf.pl –prop=./properties/**mysql**/beta/hammerdb_tproch_beta.properties

----
END OF EXAMPLE 3

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QUICK START EXAMPLE 4 - MySQL Command-Line Install
Manual Install -> Mark Active -> Run Tests Against Active Install
==

This example shows how to install MySQL manually using the direct command-line installer.

This method installs ONLY the required RPMs instead of the full 1GB MySQL bundle.

It demonstrates using the bare minimum packages and wildcard matching to install only what is needed.

IMPORTANT NOTE ABOUT SYSBENCH-LUA

If you have NOT yet built the sysbench-lua client, you must build it before running sysbench-lua tests.

Available client build actions:

```
"build-client"                          => Build test suite client
"build-client-run-tests"                => Build client and run tests
"init-start-db-build-client-run-tests" => Init and start DB, build client, run tests
"start-db-build-client-run-tests"       => Start DB, build client, run tests
```

REQUIRED MINIMAL MYSQL PACKAGES

The following RPMs are required for a complete, server-capable MySQL install:

```
mysql-community-server-8.0.45-1.el8.x86_64.rpm
mysql-community-client-8.0.45-1.el8.x86_64.rpm
mysql-community-client-plugins-8.0.45-1.el8.x86_64.rpm
mysql-community-libs-8.0.45-1.el8.x86_64.rpm
mysql-community-devel-8.0.45-1.el8.x86_64.rpm
```

These are the ONLY packages needed. TAF automatically detects and unpacks all layered RPMs when given any server-capable package.

Example directory:

```
ls ./mysql_downloads/
  mysql-community-server-8.0.45-1.el8.x86_64.rpm
  mysql-community-client-8.0.45-1.el8.x86_64.rpm
  mysql-community-client-plugins-8.0.45-1.el8.x86_64.rpm
  mysql-community-libs-8.0.45-1.el8.x86_64.rpm
  mysql-community-devel-8.0.45-1.el8.x86_64.rpm
```

COMMAND

```
perl ./taf.pl \
  --db-software-install \
  --db-software-install-packages="./mysql_downloads/mysql-community-server-8.0.45-1.el8.x86_64.rpm" \
  --verbose
```

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EXPECTED OUTPUT (CURATED AND MINIMAL)

_SelectBasePackage -> server-capable package detected (mysqld found)
PerformInstallValidations -> maker=mysql version=8.0.45

Created staging dir: /tmp/taf_unpack_xxxxx
Unpacking base package: mysql-community-server-8.0.45.rpm
Unpacking layered packages:
  mysql-community-client
  mysql-community-client-plugins
  mysql-community-devel
  mysql-community-libs

NormalizeUsrLayout:
  merging usr/bin -> bin
  merging usr/sbin -> sbin
  merging usr/lib -> lib
  merging usr/lib64 -> lib64
  merging usr/share -> share

Moving staged install to final dir:
  /opt/taf/database_software_installs/mysql-community-server-8.0.45

Active install set to:
  /opt/taf/database_software_installs/mysql-community-server-8.0.45

Install has completed. Happy benchmarking!

VERIFY ACTIVE INSTALL
----

perl ./taf.pl --list-database-software-installs

== AVAILABLE DATABASE SOFTWARE INSTALLS ===================

  1: /opt/taf/database_software_installs/mariadb-10.11.16
  2: /opt/taf/database_software_installs/mariadb-11.8.6
  3: /opt/taf/database_software_installs/mysql-9.6.0-el8.rpm-bundle
[ACTIVE] 4: /opt/taf/database_software_installs/mysql-community-server-8.0.45

RUNNING TESTS AGAINST THE ACTIVE INSTALL
----

perl taf.pl --prop=./properties/**mysql**/beta/sysbench_lua.properties
perl taf.pl --prop=./properties/**mysql**/beta/hammerdb_tprocc_beta.properties
perl taf.pl --prop=./properties/**mysql**/beta/hammerdb_tproch_beta.properties

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OPTIONAL: RUN TESTS IMMEDIATELY AFTER INSTALL

```
perl taf.pl \
  --prop=./properties/mysql/beta/sysbench_lua.properties --verbose \
  --tests=point_select \
  --threads=2 \
  --iter=1 \
  --dur=30 \
  --warmup-duration=30
```

END OF EXAMPLE 4

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END OF QUICK START
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End of Document

This Quick Start was prepared and published by the MariaDB Foundation https://mariadb.org

TAF::Perl is part of the Foundation's commitment to open, vendor-neutral tooling for the database ecosystem. The framework is maintained to support fair benchmarking, reproducible testing, and transparent engineering practices across all database makers.

For contributions, feedback, or questions, please visit the Foundation website or contact the maintainers directly.